

Random Hyperbolic Surface Spectral Rigidity — cycles 37-39

2026-05-16

Contents

Random Hyperbolic Surface Spectral Rigidity — cycles 37-39	1
Abstract	1
Introduction	2
Approach	2
Source Inventory and Timeline	2
Findings	4
Discussion	12
Open Questions	12
References	13
Appendix: Implementation Details	13

Random Hyperbolic Surface Spectral Rigidity — cycles 37-39

Abstract

Cycles 37-39 moved the campaign out of the fixed-energy local-window branch and into direct theorem-consequence work. The previous branch had been preserved as a follow-up problem in M25 because further progress required either a new localized Corollary 3.4 coefficient-variation theorem or a new noncompact trace-tail architecture. Cycle 37 therefore re-ranked the remaining extension directions and selected a conservative next target: multiplicity and spectral-cluster corollaries from Kim-Tao Theorem 1.

Cycle 38 carried out that target. It proved a deterministic transport statement: if random eigenvalues stay within a rigidity radius of deterministic reference locations, then any random spectral cluster is bounded by the number of reference locations in the corresponding expanded interval. This gives theorem-level multiplicity and cluster bounds, but the audit validated the branch as useful bookkeeping rather than a new local-statistics improvement. The bounds remain at the already understood rigidity and endpoint scales.

Cycle 39 then tested the next theorem-level input, Kim-Tao Theorem 2 on eigenfunction delocalization. It derived three deterministic corollaries: L^p interpolation, small-set mass upper bounds, and effective-support lower bounds. After an audit repair to row-level mass classification, the branch was validated as a useful theorem-level corollary package. At fixed energy, Theorem 2 excludes concentration of a normalized eigenfunction on sets below volume scale $n^{2\alpha}$ up to constants. This is distinct from M27's eigenvalue-count bookkeeping, but it is still a corollary package rather than a new proof mechanism.

No REFERENCES.md file exists in the workspace. The references section therefore lists the local paper files, session IDs, reports, proof ledgers, generated data, and figures used in this report.

Introduction

The research campaign studies Kim and Tao’s paper Eigenvalue rigidity of hyperbolic surfaces in the random cover model, using the local files 2603.01127.pdf and 2603.01127.txt as the primary source. Earlier cycles reconstructed the paper’s proof architecture, built toy and formal probes, and explored extension routes.

The immediate background for cycles 37-39 is the closure of the fixed-energy shrinking local-window route. By M25, the campaign had identified the remaining compact-support local-window problem as a precise but difficult follow-up theorem: one would need actual coefficient-variation or small- x control for the localized Corollary 3.4 numerator in the surface-group quotient family. M25 also identified a separate noncompact route, but that route would require a new trace-tail theorem. The audited decision was not to continue with another same-axis local-window empirical cycle.

The work reported here therefore asks a different question: what theorem-level consequences remain available from the already reconstructed Kim-Tao results, without solving the M25 open theorem?

The answer across these cycles is:

- M26 selected the next branch by scoring six post-local candidates.
- M27 extracted rigidity-scale multiplicity and cluster bounds from Theorem 1.
- M28 extracted L^p , small-set mass, and effective-support corollaries from Theorem 2.

Approach

The report follows the chronological sequence of the three cycles.

Cycle 37, M26, was a branch-selection cycle. It scored candidate branches by mathematical value, tractability, artifact readiness, dependence on the unresolved M25 theorem, novelty, and risk. Its role was not to prove a new theorem, but to choose the next theorem-level target with an explicit rationale.

Cycle 38, M27, used Kim-Tao Theorem 1 as the only random-cover input. Theorem 1 is the eigenvalue rigidity statement: on a high-probability event, random-cover eigenvalues remain close to deterministic reference locations. M27 converted that displacement control into interval and multiplicity bounds by deterministic counting.

Cycle 39, M28, used Kim-Tao Theorem 2 as the only random-cover input. Theorem 2 is the eigenfunction delocalization statement: for L^2 -normalized eigenfunctions below energy Λ , the sup norm is bounded by a polynomially decaying factor in the cover degree n , with explicit Λ losses. M28 converted that amplitude bound into standard norm and mass-distribution consequences.

The work was validated by cycle auditors. M27 and M28 each required one audit repair before validation: M27 repaired a decision-label issue for comparison-only rows, and M28 repaired row-level classifications for Remark 1.1 mass rows.

Source Inventory and Timeline

Cycle 37 / M26 researcher session 3d1f6457-01d1-4346-81e8-18d729d8761d. This session defined M26-post-local-extension-reprioritization. It framed the pivot after M25 and required a deterministic scoring package for at least five branches: multiplicity and spectral clusters, eigenfunction L^p /mass consequences, finite non-shrinking spectral statistics, Schreier benchmark theoremization, adjacent-model transfer, and M25-dependent local-window continuation.

Cycle 37 / M26 worker session 3aa7a047-79c7-4693-a4d5-b1b3487d5469. The worker built the M26 package: `scripts/score_post_local_extension_candidates.py`, `tests/test_post_local_extension_candidates.py`, two CSV files, two figures, the extension report, the proof-ledger attachment-point note, and a ranked follow-up list. The scorer produced six candidate rows and thirteen dependency rows. The unique recommendation was M27-multiplicity-and-cluster-corollaries-from-rigidity.

Cycle 37 / M26 auditor session 30ebba56-222e-45ef-bb37-2e1d89da85fe. The auditor validated M26 with no repair. The only minor comment was that one matrix figure used a shared color scale for positive-preference and inverse-preference axes, but the labels made the interpretation clear. The decision was VALIDATED.

Cycle 38 / M27 researcher session f07fa799-c6cf-466f-877e-0a72240a039e. This session defined M27-multiplicity-and-cluster-corollaries-from-rigidity. It asked for deterministic cluster and multiplicity consequences from Theorem 1, with separate bulk, edge, and high-energy regimes, and with an explicit test of whether the result was genuinely new or only M16 endpoint-scale bookkeeping.

Cycle 38 / M27 worker session 7d6a903e-969f-47dd-bdea-ba7d1e05f07c. The worker built `docs/proof_ledger/multiplicity_cluster_from_rigidity.md`, `reports/extension_candidates/m27_multiplicity_cluster_corollaries.md`, `reports/final/multiplicity_cluster_followup_statement.md`, the analyzer and test files, two CSV files, and two figures. The generated decision was `preserve_as_bookkeeping_corollary`.

Cycle 38 / M27 auditor session 7bfa9744-a44e-48f1-a8a8-c9229aba0b4d. The auditor found one moderate issue: `m27_cluster_regime_classification.csv` encoded `advance_multiplicity_cluster_branch` for bounded-loss hypothetical rows, even though the branch-level decision was `preserve_as_bookkeeping_corollary`. The repair forced all classification decisions to preserve the branch decision and added regression coverage rejecting any generated `advance_multiplicity_cluster_branch` row. M27 was then validated.

Cycle 39 / M28 researcher session ae184418-bd03-426c-a3ea-478d1819a0eb. This session defined M28-theorem2-lp-mass-distribution-corollaries. It asked for deterministic consequences of Theorem 2: L^p interpolation, small-set mass upper bounds, effective-support lower bounds, separation of the direct $\Lambda^{3/2}$ theorem from the Remark 1.1 $\Lambda^{1/4+\epsilon}$ variant, and explicit rejection of unsupported conclusions such as quantum ergodicity or nodal claims.

Cycle 39 / M28 worker session 968d246e-6f64-46ef-8780-93d0b9104b9d. The worker built `docs/proof_ledger/theorem2_lp_mass_corollaries.md`, `reports/extension_candidates/m28_theorem2_lp_mass_corollaries.md`, `reports/final/theorem2_followup_statement.md`, the analyzer and test files, three CSV files, and two figures. The generated decision was `advance_theorem2_consequence_branch`.

Cycle 39 / M28 auditor session 37a61811-f8e7-4403-802a-c7f26100773e. The auditor found one moderate classification issue: the mass grid labeled every Remark 1.1 mass row as `direct_theorem2_corollary`, including vacuous high-energy rows. The repair made row classification depend on the computed small-set mass exponent, effective-support exponent, and set-volume exponent. Regression tests now ensure Remark mass rows receive consequence-level labels and are not labeled `direct_theorem2_corollary`. M28 was then validated.

Findings

Finding 1: M26 Chose a Conservative Post-Local Pivot

M26’s main result was a branch decision, not a theorem. It scored six post-local candidates and selected M27 as the unique next milestone.

The ranked candidates were:

Rank	Candidate	Classification	Reason for Placement
1	Multiplicity and spectral-cluster consequences from rigidity	Kim-Tao corollary	Directly attached to Theorem 1, M2, and M16; no M25 theorem needed.
2	Eigenfunction L^p /mass-distribution consequences	Kim-Tao corollary	Directly attached to Theorem 2; likely deterministic but potentially useful.
3	Schreier benchmark theoremization	Toy/model evidence	Strong artifact readiness, but still toy-only without the M15 bridge.
4	Fixed-window non-shrinking spectral statistics	Kim-Tao corollary	Avoids shrinking-window obstruction but risks repackaging global rigidity.
5	Adjacent-model transfer	Requires new theorem	High value but too model-dependent for one cycle.
6	M25-dependent shrinking local-window continuation	Requires new theorem	Immediate next step remains the unresolved coefficient-variation or trace-tail theorem.

The score table is stored in `data/extension_candidates/post_local_extension_candidate_scores.csv`, and the dependency table is stored in `data/extension_candidates/post_local_extension_candidate_dependencies.csv`.

The audited decision was that this was a defensible pivot. The M25-dependent local-window branch remained preserved as a future problem, but it was explicitly deprioritized for immediate continuation.

Finding 2: M27 Converted Rigidity Into Cluster Bounds, But Only at the Rigidity Scale

M27 derived a deterministic transport lemma from Theorem 1. Let μ_j denote random-cover eigenvalues and λ_j denote deterministic reference locations. If, on a high-probability event,

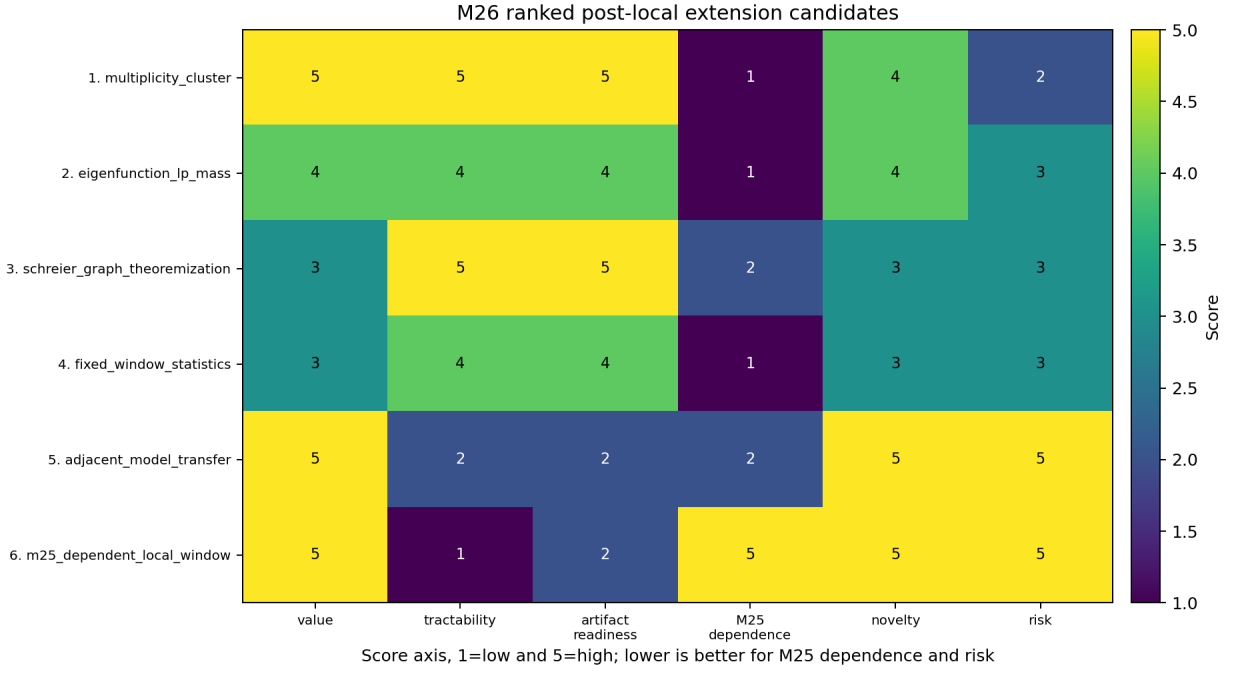


Figure 1: Ranked post-local extension candidates by value, tractability, dependency risk, and artifact readiness. The M25-dependent local-window branch is preserved but ranked last because its immediate next step is still an open theorem.

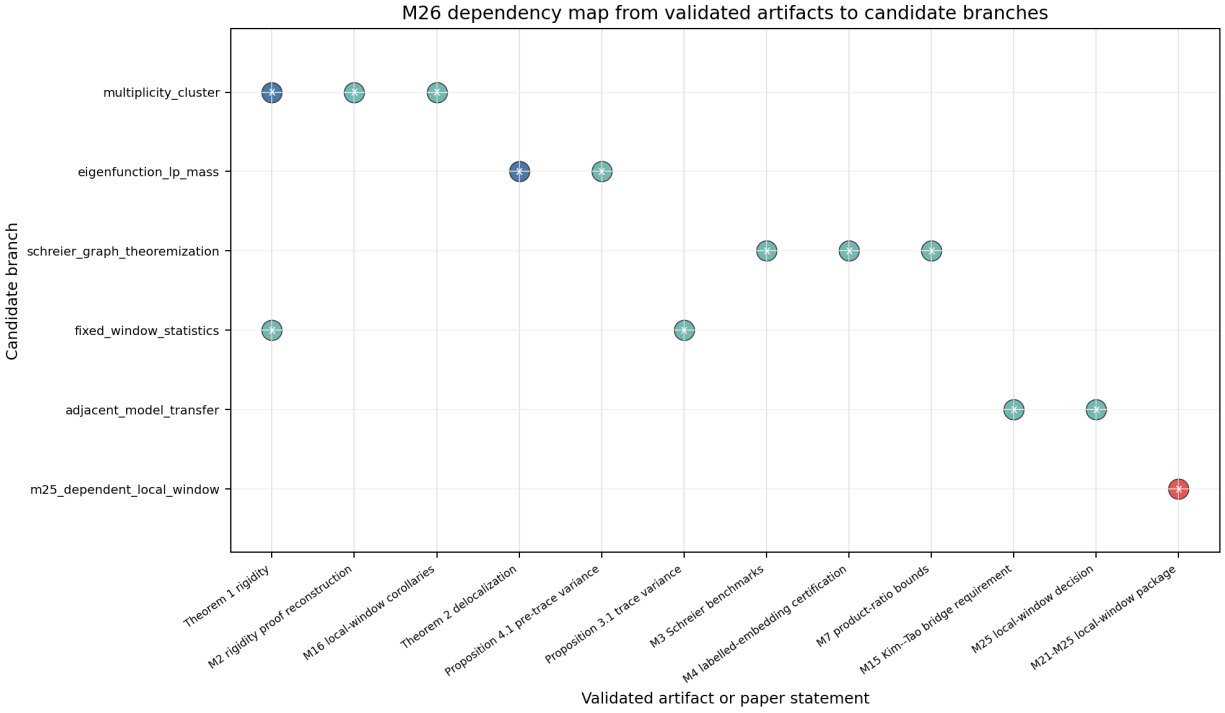


Figure 2: Dependency map from validated Kim-Tao proof artifacts and Phase II results to post-local candidate branches. M27 attaches directly to Theorem 1 and prior rigidity/Weyl reconstruction.

$$|\mu_j - \lambda_j| \leq R_j,$$

then any interval I can contain only those random eigenvalues whose reference locations lie in an expanded interval. Using a uniform radius R on the relevant spectral range,

$$\#\{j : \mu_j \in I\} \leq \#\{j : \lambda_j \in I^{+R}\}.$$

For a single point, this becomes an exact multiplicity envelope:

$$\text{mult}_{X_n}(\lambda) \leq \#\{j : |\lambda_j - \lambda| \leq R\}.$$

For Kim-Tao Theorem 1, the rigidity radius has the form

$$R = C_\epsilon \Lambda_{\max}^{1/2+\epsilon} n^{-\alpha_R}.$$

The deterministic reference locations are defined by

$$F(\lambda_j) = \frac{j}{(2g-2)n}, \quad F(\Lambda) = \int_0^{\sqrt{\Lambda-1/4}} r \tanh(\pi r) dr.$$

In the fixed bulk, where $\Lambda_0 > 1/4$,

$$F'(\Lambda_0) = \frac{1}{2} \tanh\left(\pi \sqrt{\Lambda_0 - 1/4}\right),$$

so the zero-width multiplicity envelope becomes

$$1 + 2(2g-2)nF'(\lambda)C_\epsilon \lambda^{1/2+\epsilon} n^{-\alpha_R} + O(nR^2).$$

At the spectral edge, the bulk density approximation fails because $F'(1/4) = 0$. The correct expansion is

$$F(1/4 + \Delta) = \frac{\pi}{3} \Delta^{3/2} + O(\Delta^{5/2}),$$

which gives an edge envelope of the form

$$1 + (2g-2)n\frac{\pi}{3}(\Delta + R)^{3/2} + O(n(\Delta + R)^{5/2}).$$

At high energy, $F'(\Lambda) \rightarrow 1/2$, while the radius grows like $\Lambda^{1/2+\epsilon} n^{-\alpha_R}$. The high-energy count is therefore dominated by the rigidity radius, not improved by density decay.

The generated classification table had 12 rows after audit repair. At $n = 10^8$, representative proved-shape rows included:

Regime	Classification	Decision
Edge Δ at the rigidity scale	edge_endpoint_equivalent	preserve_as_bookkeeping_corollary
Near fixed edge	rigidity_scale_cluster_bound	preserve_as_bookkeeping_corollary
Fixed bulk	tautological_or_endpoint_only	preserve_as_bookkeeping_corollary
High energy	high_energy_loss_dominated	preserve_as_bookkeeping_corollary

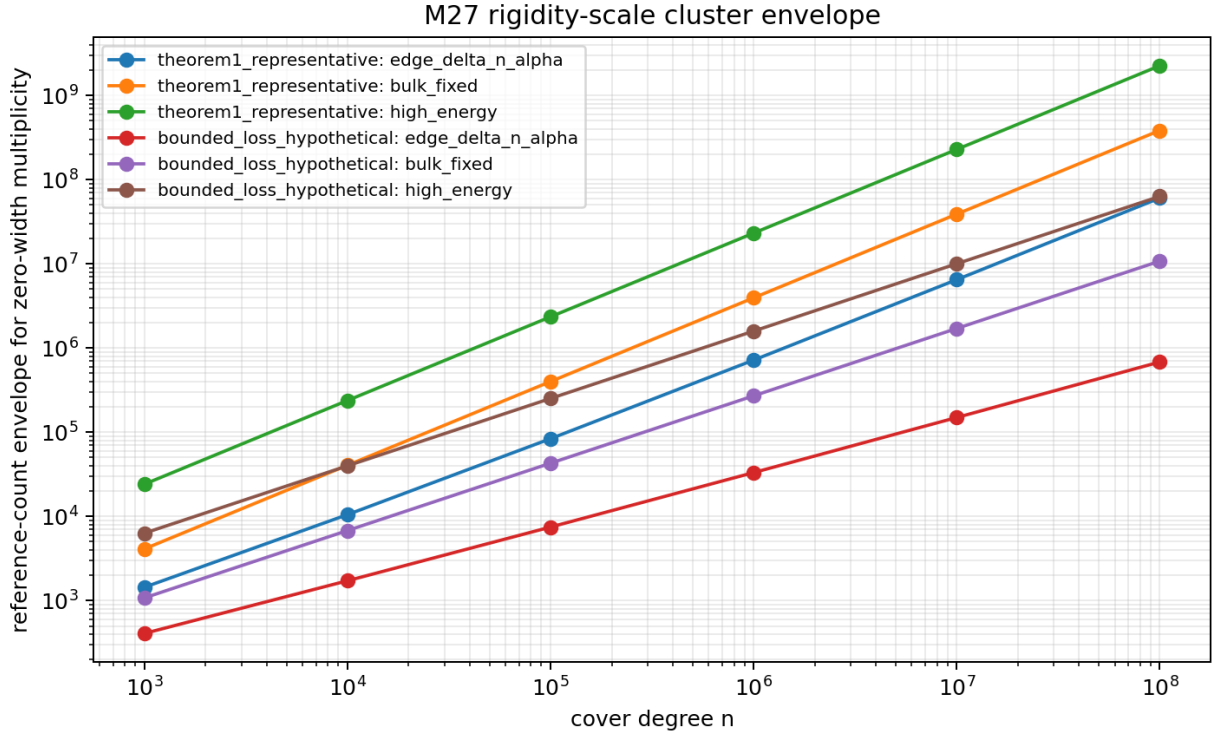


Figure 3: Log-log cluster-envelope scaling versus cover degree for bulk, edge, and high-energy representative energies. The envelopes remain at the rigidity scale for proved-shape representative exponents.

The audited conclusion was that M27 is mathematically correct and theorem-level, but not a new local-statistics result. It bounds clusters and exact multiplicities by deterministic reference locations in a rigidity-expanded interval. It does not imply simplicity, level repulsion, microscopic statistics, or endpoint-beating local counts.

Finding 3: M28 Produced a Theorem 2 Mass-Support Corollary Package

M28 turned to Kim-Tao Theorem 2, which controls eigenfunction amplitudes. For an L^2 -normalized eigenfunction u_j with eigenvalue below Λ , the direct Theorem 2 input has the form

$$\|u_j\|_\infty \leq C\Lambda^{3/2}n^{-\alpha}.$$

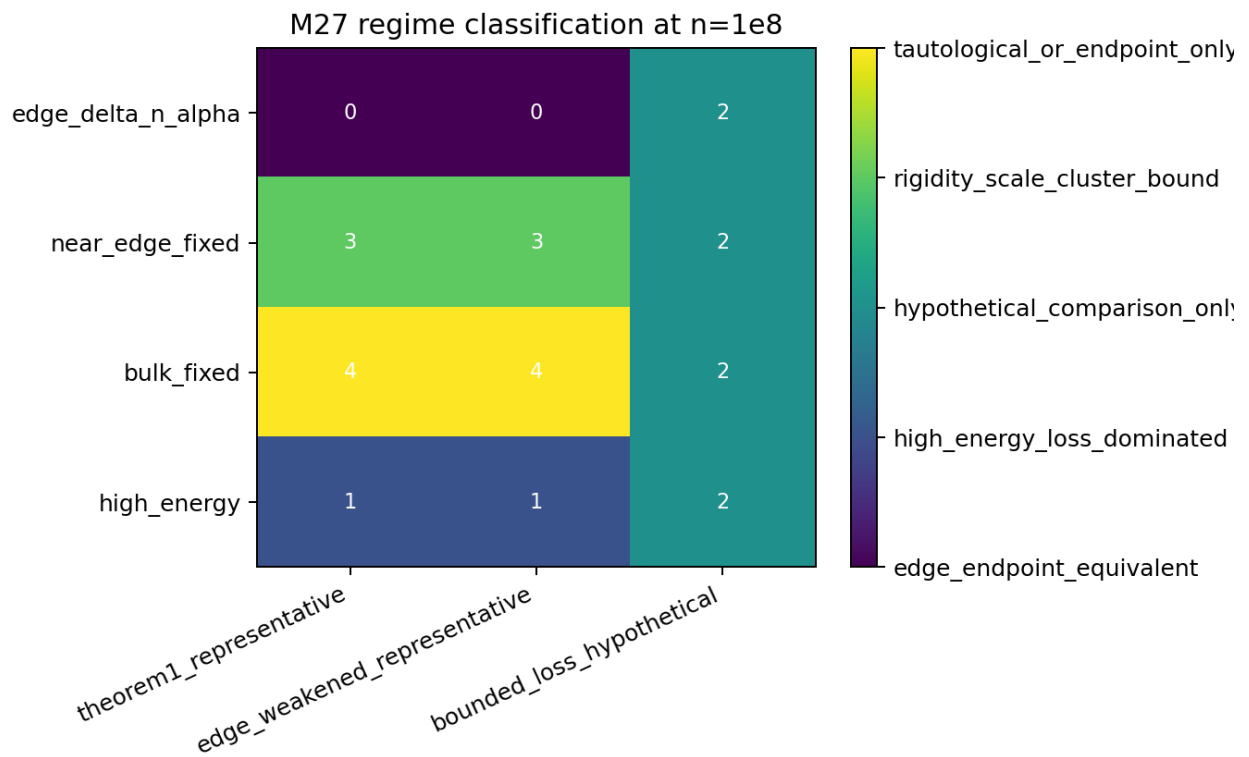


Figure 4: Regime classification map at fixed representative cover degree. The post-audit decision rows preserve the branch-level decision rather than advancing comparison-only hypothetical rows.

Remark 1.1 gives a different high-energy tradeoff:

$$\|u_j\|_\infty \leq C_\epsilon \Lambda^{1/4+\epsilon} n^{-\alpha'}.$$

M28 kept these two models separate because the improved Λ exponent comes with a different n exponent and constant.

Let $M_{\Lambda,n}$ denote either admissible amplitude envelope. Three deterministic consequences follow.

First, interpolation between L^2 and L^∞ gives, for $2 \leq p \leq \infty$,

$$\|u\|_p \leq \|u\|_\infty^{1-2/p} \|u\|_2^{2/p} \leq M_{\Lambda,n}^{1-2/p}.$$

The endpoint checks are exact: at $p = 2$ this recovers $\|u\|_2 = 1$, and at $p = \infty$ it recovers Theorem 2.

Second, for any measurable set $A \subset X_n$,

$$\int_A |u|^2 \leq \|u\|_\infty^2 \text{vol}(A) \leq M_{\Lambda,n}^2 \text{vol}(A).$$

This is nontrivial relative to the identity $\int_A |u|^2 \leq 1$ when

$$\text{vol}(A) < M_{\Lambda,n}^{-2}.$$

Third, if a measurable set E carries mass θ , then

$$\theta \leq \int_E |u|^2 \leq M_{\Lambda,n}^2 \text{vol}(E),$$

so

$$\text{vol}(E) \geq \theta M_{\Lambda,n}^{-2}.$$

At fixed energy in the direct Theorem 2 model, this gives

$$\text{vol}(E) \gtrsim_\Lambda \theta n^{2\alpha}.$$

This is the main preserved consequence of M28: fixed-energy eigenfunctions cannot concentrate mass on sets of volume $o(n^{2\alpha})$, up to constants. Since the cover volume is order n , this is partial delocalization rather than equidistribution.

The generated classification file contains six rows:

Item	Classification	Meaning
Theorem 2 sup-norm input	direct_theorem2_corollary	The paper's input statement or Remark 1.1.

Item	Classification	Meaning
L^p interpolation	standard_interpolation	Deterministic norm interpolation.
Small-set mass	nontrivial_mass_delocalization	Fixed-energy exclusion of concentration below polynomial volume.
High-energy limitations	bookkeeping_only	Λ growth can erase the n gain.
Quantum ergodicity/equidistribution claims	unsupported_stronger_claim	Sup-norm control alone does not force mass into specified regions.
Branch decision	nontrivial_mass_delocalization	Preserve the Theorem 2 consequence package.

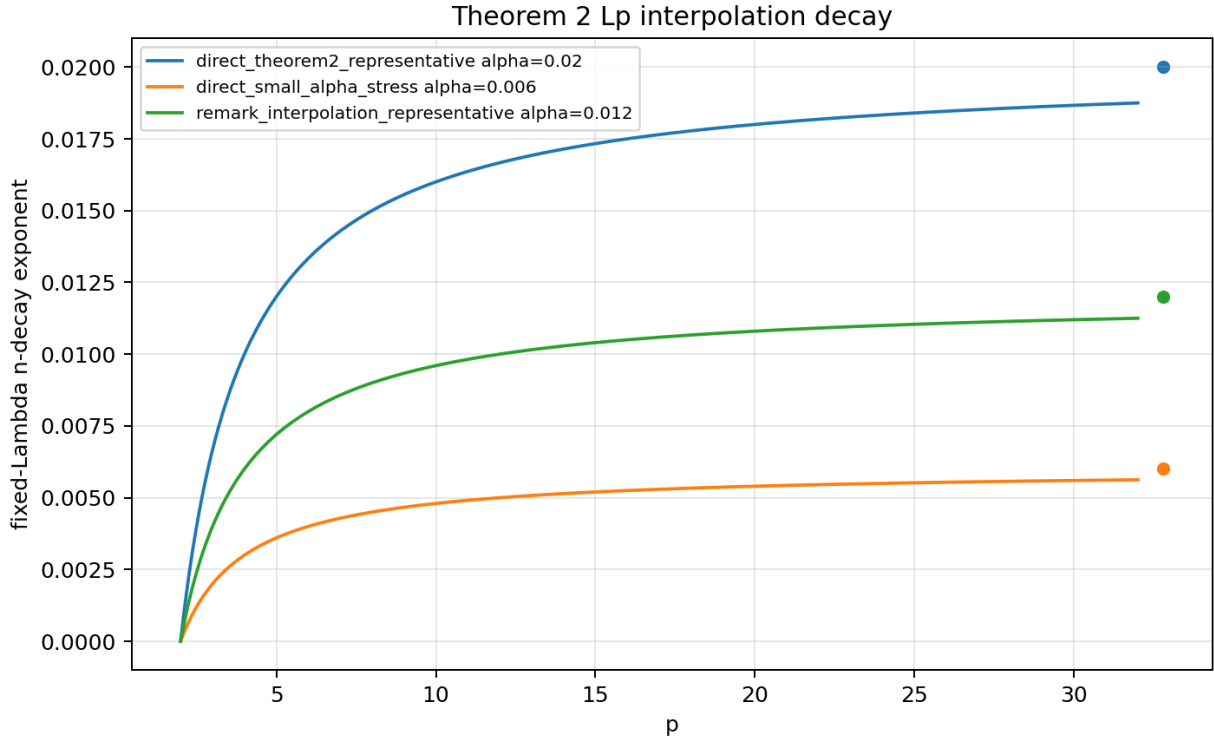


Figure 5: Decay exponent for L^p interpolation as a function of p . The exponent vanishes at $p = 2$ and recovers the Theorem 2 exponent at $p = \infty$.

The audit repair mattered for downstream planning. Before repair, every Remark 1.1 mass row was classified as `direct_theorem2_corollary`, including vacuous high-energy rows. After repair, row labels depend on the computed mass exponent, effective-support exponent, and set-volume exponent. This prevents the report from presenting high-energy bookkeeping rows as direct theorem-level mass consequences.

The validated decision was `advance_theorem2_consequence_branch`, with a precise limitation: this is a corollary package, not a new proof mechanism.

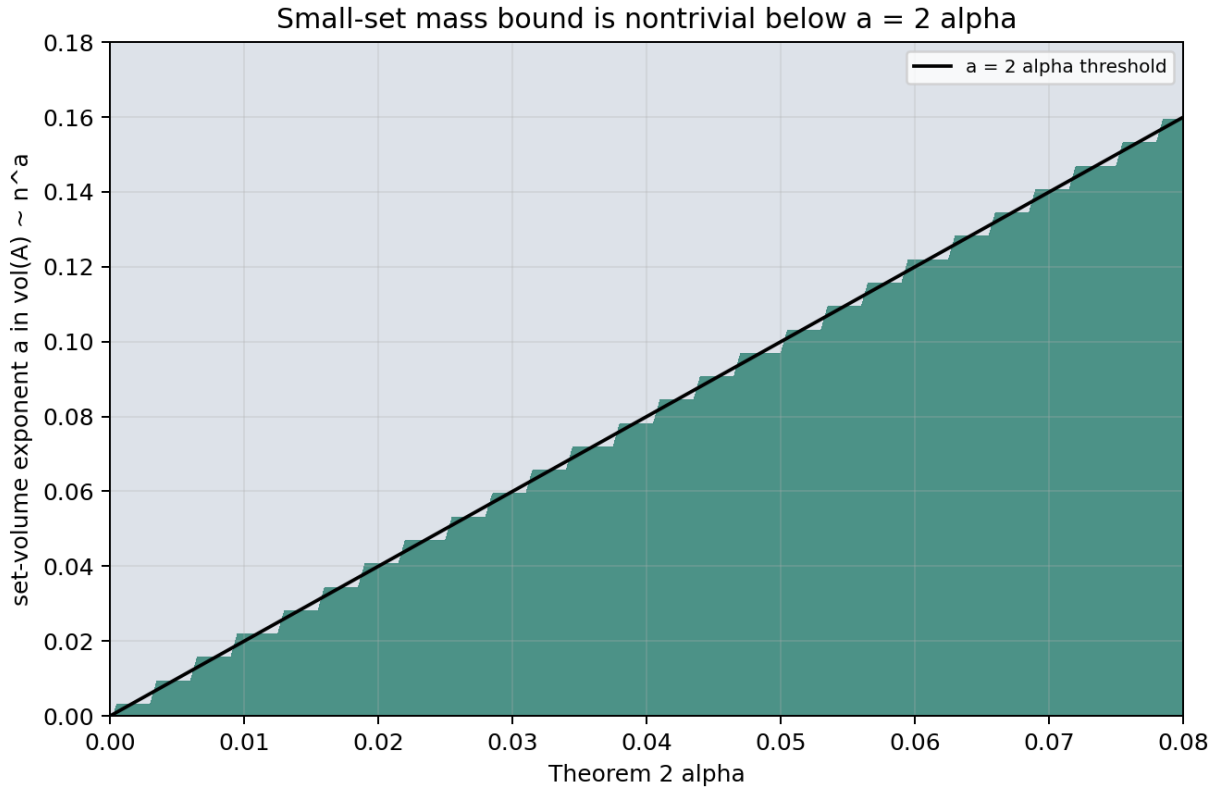


Figure 6: Set-volume exponent phase diagram for the small-set mass envelope. The nontrivial region is where $M_{\Lambda,n}^2 \text{vol}(A) < 1$.

Discussion

The three cycles form a clean post-local sequence.

M26 prevented the campaign from drifting back into the local-window obstruction. It did this by scoring candidate branches and making the M25 dependency explicit. The best immediate target was not the most ambitious one; it was the one with the strongest attachment to validated proof artifacts and the least dependence on unresolved coefficient-variation input.

M27 then tested that target. The resulting multiplicity and cluster bounds are useful because they package Theorem 1 into a direct statement about observed random-cover spectral clusters. But the audit-confirmed branch decision is conservative: the bounds are transport bounds at the rigidity radius. They do not create new local spectral statistics.

M28 produced a stronger preserved consequence because Theorem 2 controls eigenfunction amplitude directly. The L^p inequalities are standard interpolation, but the effective-support statement has a clear random-cover interpretation. At fixed energy, a normalized eigenfunction cannot place positive mass on a set much smaller than $n^{2\alpha}$ in volume. This is weaker than full-volume equidistribution but stronger in nature than M27's eigenvalue-count bookkeeping.

The cumulative lesson is that direct theorem-consequence branches are valuable for extracting clean corollaries and follow-up statements, but they should not be repeated indefinitely. After M27 and M28, the next high-value step should either exploit the internal pre-trace local-mass step behind Theorem 2 in a carefully scoped way, or pivot to a less bookkeeping-heavy direction such as finite non-shrinking spectral statistics or Schreier benchmark theoremization.

Open Questions

1. **Can the Theorem 2 pre-trace local-mass intermediate be made into a standalone fixed-domain spatial theorem?** M28 records an internal proof-ledger statement for fundamental-domain cutoffs, but it did not elevate that to arbitrary moving sets or all balls. Doing so would require reopening the Proposition 4.1 cutoff architecture.
2. **Can finite non-shrinking spectral statistics produce more than global rigidity repackaging?** M26 ranked this below M27 and M28, but it remains a possible next branch if centered fixed-width statistics can avoid the shrinking-window support obstruction.
3. **Can the Schreier/random-regular benchmark branch be theoremized without overclaiming transfer to Kim-Tao?** M26 kept this branch viable as a toy/model theoremization path, but M15's bridge obstruction still blocks direct transfer to actual surface-group quotient families.
4. **Can the M25 local-window follow-up theorem be attacked directly?** The compact route requires localized Corollary 3.4 coefficient-variation or small- x control for the actual folded surface-group quotient family. The noncompact route requires a trace-tail theorem with sufficient tail-rate dominance. Neither was pursued in cycles 37-39.
5. **What is the correct stopping point for direct corollary extraction?** M27 and M28 show that direct consequences are worth recording, but the campaign should avoid another cycle that only reclassifies deterministic interpolation or endpoint bookkeeping unless a genuinely new local-mass statement is extracted.

References

No REFERENCES.md file was present in the workspace during this reporting pass. The following local sources were used.

- Kim-Tao paper files: 2603.01127.pdf, 2603.01127.txt.
- Cycle 37 sessions: researcher 3d1f6457-01d1-4346-81e8-18d729d8761d, worker 3aa7a047-79c7-4693-a4d5-b1b3487d5469, auditor 30ebba56-222e-45ef-bb37-2e1d89da85fe.
- Cycle 38 sessions: researcher f07fa799-c6cf-466f-877e-0a72240a039e, worker 7d6a903e-969f-47dd-bdea-ba7d1e05f07c, auditor 7bfa9744-a44e-48f1-a8a8-c9229aba0b4d.
- Cycle 39 sessions: researcher ae184418-bd03-426c-a3ea-478d1819a0eb, worker 968d246e-6f64-46ef-8780-93d0b9104b9d, auditor 37a61811-f8e7-4403-802a-c7f26100773e.
- M26 artifacts: reports/extension_candidates/m26_post_local_extension_reprioritization.md, docs/proof_ledger/post_local_branch_attachment_points.md, reports/final/post_local_followup_ranked_problem_list.md, data/extension_candidates/post_local_extension_candidate_scores.csv, data/extension_candidates/post_local_extension_candidate_dependencies.csv.
- M27 artifacts: reports/extension_candidates/m27_multiplicity_cluster_corollaries.md, docs/proof_ledger/multiplicity_cluster_from_rigidity.md, reports/final/multiplicity_cluster_followup_statement.md, data/extension_candidates/m27_cluster_bound_grid.csv, data/extension_candidates/m27_cluster_regime_classification.csv.
- M28 artifacts: reports/extension_candidates/m28_theorem2_lp_mass_corollaries.md, docs/proof_ledger/theorem2_lp_mass_corollaries.md, reports/final/theorem2_followup_statement.md, data/extension_candidates/m28_lp_bound_grid.csv, data/extension_candidates/m28_mass_distribution_grid.csv, data/extension_candidates/m28_corollary_classification.csv.

Appendix: Implementation Details

Code Organization

New or cycle-relevant scripts:

File	Lines	Purpose
scripts/score_post_local_extension_candidates.py	324	Scores post-local extension candidates after M25.
scripts/analyze_multiplicity_cluster_bounds.py	240	Computes M27 rigidity-scale cluster and multiplicity envelopes.
scripts/analyze_theorem2_lp_mass_corollaries.py	247	Computes M28 L^p , small-set mass, and support-corollary grids.

New or cycle-relevant tests:

File	Lines	Purpose
tests/test_post_local_extension_candidates.py	72	Validates M26 candidate coverage, ranking, and unique recommendation.

File	Lines	Purpose
tests/test_multiplicity_ cluster_bounds.py	72	Validates M27 formulas, edge scaling, monotonicity, and repaired branch decision.
tests/test_theorem2_lp_ mass_corollaries.py	83	Validates M28 interpolation endpoints, mass thresholds, unsupported-claim exclusions, and repaired Remark-row classifications.

Generated Data

File	Rows Excluding Header	Notes
data/extension_ candidates/post_local_ extension_candidate_ scores.csv	6	M26 ranked candidates.
data/extension_ candidates/post_local_ extension_candidate_ dependencies.csv	13	M26 dependency map rows.
data/extension_ candidates/m27_cluster_ bound_grid.csv	72	M27 cluster-envelope grid.
data/extension_ candidates/m27_cluster_ regime_classification.csv	12	M27 regime classification after audit repair.
data/extension_ candidates/m28_lp_bound_ grid.csv	648	M28 L^p interpolation grid.
data/extension_ candidates/m28_mass_ distribution_grid.csv	96	M28 mass-distribution grid after audit repair.
data/extension_ candidates/m28_corollary_ classification.csv	6	M28 corollary and branch classifications.

Figure Inventory

Figure	Dimensions	Used For
reports/figures/m26_post_ local_candidate_matrix.png	1872x1044	M26 candidate scoring matrix.

Figure	Dimensions	Used For
reports/figures/m26_extension_dependency_map.png	2015x1188	M26 dependency map.
reports/figures/m27_cluster_bound_vs_n.png	1440x900	M27 cluster envelope versus n .
reports/figures/m27_regime_phase_diagram.png	1350x864	M27 regime classification.
reports/figures/m28_lp_decay_by_p.png	1440x900	M28 L^p decay exponents.
reports/figures/m28_mass_scale_phase_diagram.png	1350x900	M28 mass-scale phase diagram.

Validation Results

The final validation commands run during this reporter pass were:

```
python3 -m long_exposure.tools.promise_check .
python3 -m long_exposure.tools.org_check .
```

promise_check exited successfully and reported:

```
events: 110, plan milestones: 28
```

Its warnings were historical: one old noncanonical docs/paper_map/ path and orphan prior cycle reports under reports/cycles/.

org_check exited successfully. Its warnings were historical root files and older figures under docs/.

The reporter also updated MANIFEST.md after gathering. The updated manifest has 156 lines and records the workspace snapshot through M28.

Manifest Snapshot

After cycles 37-39, the manifest records:

Category	Count
Python scripts	37 files, 10,302 lines
Python tests	29 files, 2,376 lines
Canonical CSV datasets	74
PNG figures under reports/figures/	67
Markdown/DOT/PNG documentation and report artifacts	159
Promise ledger events	110
Plan milestones	28

There was no ## Key Files section in the previous manifest, so no protected final-reporter section needed preservation.

Cross-Reference Map

Origin	Consuming Artifact	Role
M25 local-window branch decision	M26 candidate scorer	Forces pivot away from same-axis local-window work.
M26 score table	M27 researcher brief	Selects multiplicity and cluster corollaries as next branch.
Kim-Tao Theorem 1 / M2 / M16	M27 proof ledger and analyzer	Supplies rigidity radius and reference-density formulas.
M27 audit repair	M27 classification table and tests	Prevents comparison-only rows from advancing the branch.
M27 branch decision	M28 researcher brief	Moves from eigenvalue bookkeeping to eigenfunction delocalization consequences.
Kim-Tao Theorem 2 / M2 delocalization ledger	M28 proof ledger and analyzer	Supplies sup-norm input for L^p , mass, and support corollaries.
M28 audit repair	M28 mass grid and tests	Prevents vacuous Remark rows from being labeled direct theorem corollaries.
M28 final statement	Future branch planning	Preserves Theorem 2 mass-support corollary and points to pre-trace local-mass refinement or a new pivot.